
SCIE30001 - SCIENCE RESEARCH PROJECT

**All-sky Simulation of SkyHopper's
Exoplanet Transit Detection**

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1 INTRODUCTION

1.1 SkyHopper and its Vision

The SkyHopper is a 12U, near-infrared (NIR) cubesat designed with a major objective to identify the earliest Gamma Ray Burst (GRB) afterglows at the furthest observable surface of the universe. However, GRB observation highly condition dependent. Thus during the rest of its time SkyHopper will be used to:

- a. search for earth-like planets in the habitable zone through planetary transits
- b. explore the time-variable universe (e.g. identifying transient phenomena)
- c. measure the Cosmic Microwave Background (CMB) radiation
- d. unveil planetary system formation

Although SkyHopper covers a whole new discovery space in search of exoplanets, part of it coincides with TESS (Transiting Exoplanet Searching Satellite) as shown in the Figure 1.1. Hence, SkyHopper is not just capable of searching for faraway planets of longer orbital periods, it can also follow-up on TESS' science targets to check for transit timing variations.^[1]

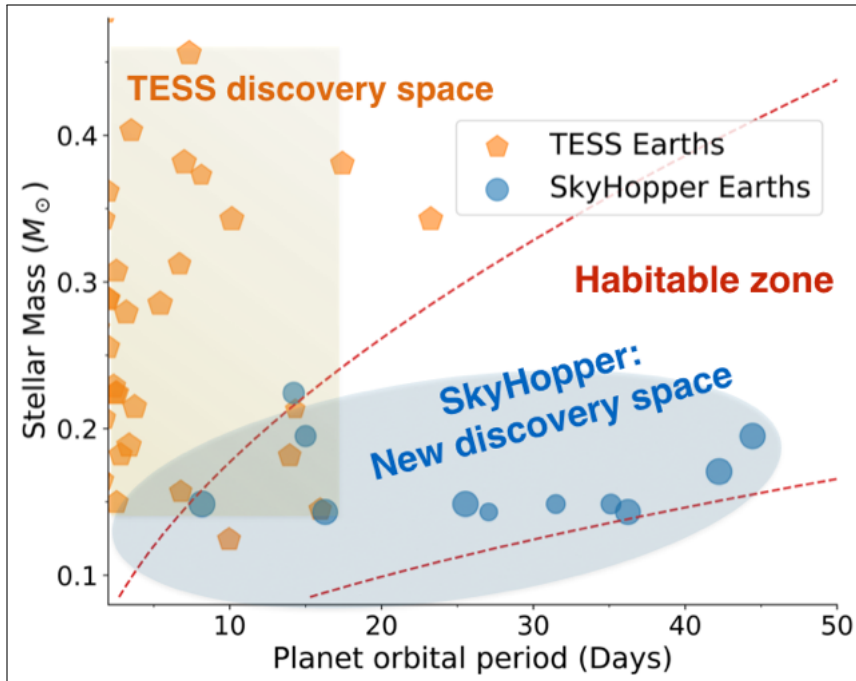


Figure 1.1: SkyHopper and TESS discovery space. (Image credits to Prof. Michele Trenti)

1.2 Project Overview

This science research project focussed on the objective of searching for exoplanets with relatively short orbital periods ($\sim 5 - 10$ days) around known low mass stars ($\sim 0.2 - 0.4M_{\odot}$) using SkyHopper. This project aimed to simulate the SkyHopper's exoplanet transit detection in "python". Through the simulation, the efficiency of SkyHopper and its probability of finding exoplanet transits can be determined. These results will be useful for optimising the search for exoplanets.

The simulation written this project is an improved version of the preliminary simulation I wrote over my winter project in 2018. In this report, I will be giving an detailed overview on methods used in the preliminary single-coordinate simulation , and how it was expanded into the all-sky simulation. I will also mention briefly where this simulation can lead to at the end of the report to conclude the project.

NOTE: This report (especially Sections 2, 3 and 4) contains information taken from my [winter research report](#) as they are crucial information that builds towards the improved simulation for the science research project.

2 PLANETARY TRANSIT

Searching for exoplanets using the transit method is only possible for planets with orbital planes approximately parallel to our line of sight. For simplicity, the simulation was developed under the assumption that all the planets observed satisfy this condition.

A transit happens when a planet orbiting a host star passes between the star and the observer along the line of sight of the observer. Having the planet partially blocking the light of the star from reaching the observer (telescope), there will a significant reduction in the flux measured.

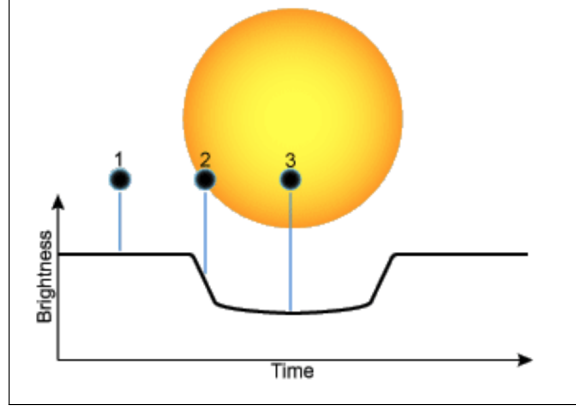


Figure 2.1: Light curve of a star during planetary transit with 1, 2 and 3 indicating position of the planet with respect to the points on the light curve. (Credits: STARE Project^[2])

However, we have to be very careful in identifying these light curve dips. Some observed dips could possibly be false signals caused by instrumental errors or non-planetary bodies passing by the front of the host star, but if the dip is consistently observed at regular periods, the object is most likely a planet.

3 SIMULATION INPUT

3.1 SkyHopper Visibility

For approximately half of SkyHopper's orbital time (~ 96 min), its vision on that specific patch of the sky is occulted by the Earth itself. Thus, SkyHopper cannot continuously stare at the same patch of the sky.

To take that into account, one of the most crucial inputs of the simulation will be the all-sky visibility map of SkyHopper provided by Robert Mearns. This visibility map provides information on the visibility of SkyHopper on every location across the sky at any time of the year.

The visibility map provided is a 3-dimensional array: [timestep, Ra, Dec], with each timestep representing a 6 minute cadence. "Ra" and "Dec" stands for Right Ascension and Declination respectively and they are used to denote the location in the sky. While the visibility of SkyHopper in this map is given in boolean form: "True" for visible and "False" for not visible, it is easier to visualise plotting in a numerical axes. To do that, the boolean can be converted into floats such that True = 1 and False = 0 which can then be plotted as shown in Figure 3.1.

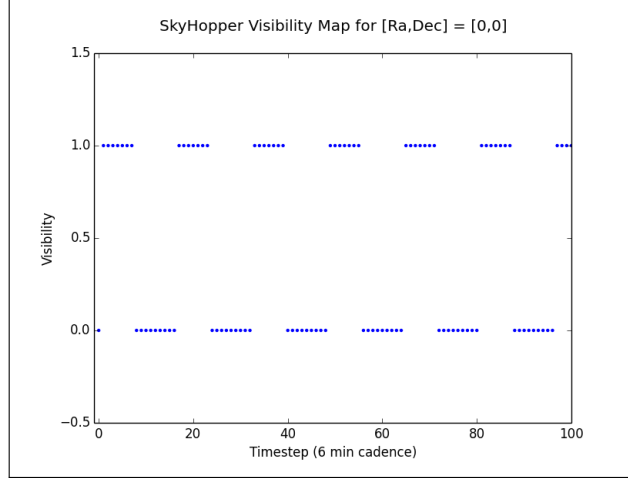


Figure 3.1: SkyHopper visibility map of $[Ra, Dec]=[0,0]$ for the first 100 timesteps (i.e. first 600 minutes) of the year where each plot point represents one discrete 6-minute timestep.

3.2 Parameter File

This simulation is capable of exploring exoplanet transit detection efficiency under the influence of two parameters: the *stellar mass to solar mass ratio*, M , and *orbital period in days*, T . These are the first two user inputs.

The next input would be the duration of SkyHopper observing a specific patch of the sky. This parameter will affect the efficiency of transit detection. For example, to observe the transit of a planet with 7-day orbital period, SkyHopper will have to be pointing at the target for a sufficiently long time. It is obvious that the longer SkyHopper stares at a target, the more likely it would find a transit. However, we do not want to waste too much of SkyHopper’s observation time on the same science target. Hence the observation time has to be chosen wisely. Here, I have chosen the observation time to be 4 times the orbital period of the planet.

Then there is the number of runs of the simulation. It is vague what this means at this point, but it will make more sense after the discussion on the methods of the simulation, so I will come back to explain this input later.

Finally, we need to consider at which time of the year will SkyHopper be observing the target. Since this is an uncertain parameter, the input must include a random number generator that could provide a random timestep at which SkyHopper starts observing. Since there are 87,600 timesteps (each of 6-minute cadence) in one year, the random number generator is set to provide a random integer in the range 0 to 87600, inclusive. If the number generated is (let’s say) on the last day of the year, there is a section in the simulation that would take that into account and reset the visibility map to the start of a new year. (However, if the user knows when exactly SkyHopper will be observing, this parameter can also be provided manually).

To sum that up, the parameter file should look like this and should be written in the format “.py” to be compatible with the python script of the simulation:

```

M_frac = 0.20
# Ratio of stellar mass to solar mass, M

T_days = 7
# Orbital Period in days, T

tg_days = 28
# Observation time in days (User input)

run_number = 3
# Number of runs

tg_start = random.randint(0, 87600)
# Random number generator

```

3.3 Variables and Derived Quantities

Many other important quantities required for the simulation can be derived through the two input parameters M and T . These *derived quantities* include^[3]:

a. **Stellar Radius, R :**

Given the stellar mass-radius relationship

$$R = kM^x \quad (1)$$

Assuming that all the stars within the parameter space are main sequence stars, we have $k = 1$ and $x = 0.8$. Hence the stellar radius is:

$$R = M^{0.8} \left(\frac{R_{\odot}}{M_{\odot}^{0.8}} \right) \quad (2)$$

b. **Semi-major axis, a :**

By Kepler's Third Law of Planetary Motion

$$a = \left(\frac{T^2 G (M + m)}{4\pi^2} \right)^{\frac{1}{3}} \quad (3)$$

where m = mass of planet. However, since $m \ll M$, we can assume that

$$a \approx \left(\frac{T^2 G M}{4\pi^2} \right)^{\frac{1}{3}} \quad (4)$$

c. **Planetary Transit Duration, t :**

$$t = 13 \left(\frac{M}{M_{\odot}} \right)^{-\frac{1}{2}} \left(\frac{a}{1AU} \right)^{\frac{1}{2}} \left(\frac{R}{R_{\odot}} \right) \text{ hours} \quad (5)$$

where $M_{\odot}$ = Solar mass, $R_{\odot}$ = Solar radius, G = Gravitational Constant

4 SINGLE COORDINATE SIMULATION

4.1 Defining Planetary Transit Function

The planetary transit duration derived from a given M and T in Equation 5 can be used to define the light curve of the host star. For simplicity, the gradual flux change during a planetary transit is seen in point "2" of Figure 2.1. The light curve here will be assuming a direct drop in luminosity at the start of a planetary transit.

Firstly, we introduce a new dimensionless parameter " x_{planet} " which represents the location of the planet in its orbit. In this simulation, the start of a new planetary orbit is to be at the instant where a transit starts with $x_{planet} = 0$. Then the location of the planet 6 minutes after the start of a transit will be $x_{planet} = 0$ and so on. Hence, given that:

$$x_{full} = \frac{T}{6} \quad (6)$$

the location of the planet at the end of its orbit (before the next transit) is $x_{planet} = x_{full}$.

A real light curve is a luminosity vs. time plot which we are unable to produce here due to the lack of data on the actual brightness of the star. Thus, this simulation uses a "Planetary Transit Function" (PTF) which is a piecewise continuous step function analogous to the light curve with a luminosity dip:

$$p(x_{planet}) = \begin{cases} 0.5 & x_{planet} \leq x_{transit} \\ 0 & x_{transit} < x_{planet} \leq x_{full} \end{cases} \quad (7)$$

where $x_{transit} = t/6$ is the number of timesteps per planetary transit.

In words, $p(x_{planet}) = 0.5$ is when the planet is in transit and $p(x_{planet}) = 0$ when the planet is not in transit. Plotting orbit of the $p(x_{planet})$ against x_{planet} gives the synthetic light curve of one planetary orbit as shown in Figure 4.1.

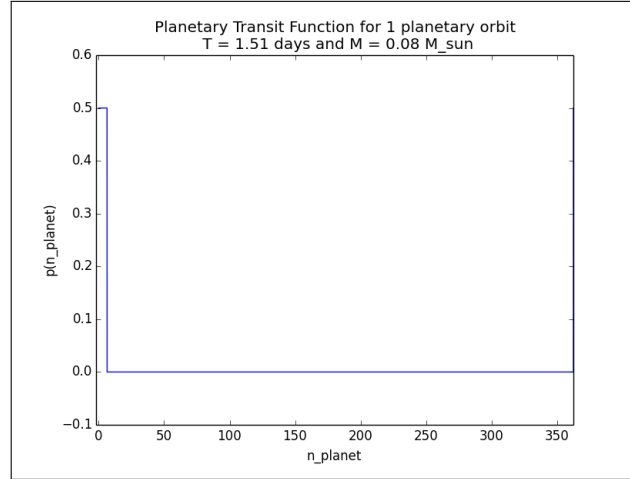


Figure 4.1: Planetary Transit Function for 1 full planetary orbit.

Combining many single-orbital light curve together as shown in Figure 4.2 provides the synthetic light curve of a planet over a long period of time (i.e. on the scale of SkyHopper's observation period). The spike(s) in the figures represent transits. (One spike in Figure 4.2 is circled for illustration.)

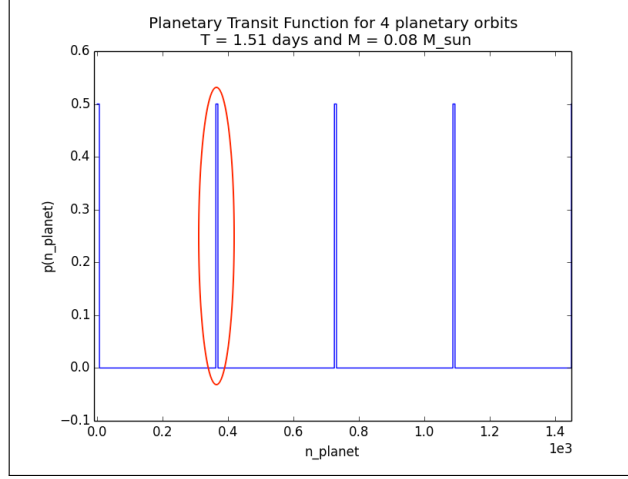


Figure 4.2: Planetary Transit Function for 4 full planetary orbits.

4.2 Converting Planetary Transit Function into an Array

The PTF and SkyHopper’s visibility map can be combined to provide useful information on SkyHopper’s exoplanet detection efficiency. However, the PTF is a continuous function while the visibility map is an array.

The visibility array of a single [Ra, Dec] consists of $N = (tg_days \times 24 \times 60)/6$ elements, each of which represents one timestep and carry the value 0 or 1 which denotes the visibility of SkyHopper as previously mentioned. Thus, the continuous PTF has to be discretised and converted into an array of also N elements so that the shape of the array matches the visibility map. Elements in this array carry the value 0 or 0.5 depending on the location of planet in its orbit and the first element in the PTF array will depend on the planet’s location in orbit at the start of observation.

4.3 Planet Location in Orbit at the Start of Observation

Since we are uncertain about the planet’s orbital location at the start of SkyHopper’s observation, it is essential to have another random number generator to provide a random position of the planet in the orbit at the start of SkyHopper’s observation.

However, unlike the observation start timestep generator in the parameter file, this random number generator does not only provide integers. It can generate any number up to 8 decimal places between 0 and N_{planet} . The reason being that the orbit of the planet around the star is continuous and the planet could be anywhere in this orbit at the start of observation.

This parameter will affect PTF array that will be combined with the visibility map of SkyHopper as mentioned in Section 4.2. Due to such randomness and its significant effect on SkyHopper’s efficiency in detecting exoplanets, a Monte Carlo simulation is required to obtain an average transit detection efficiency of SkyHopper. This will be discussed in more details later in Section 6.

4.4 Combining Visibility Map and PTF Array

With the random orbit location generator, the shifted PTF array can now be combined with the visibility map of a single [Ra, Dec] over the selected period of time by addition. Each element of the resulting N-shaped array can take one of four different possible values.

- 1.5 $\implies$ Transit is detected
- 1.0 $\implies$ Non-transit is detected
- 0.5 $\implies$ Transit is missed
- 0.0 $\implies$ Non-transit is missed

4.5 Visualising Transit Detection

For better understanding of how the simulation works, here are two plots, each of which has both the visibility map and PTF array plotted against observation timestep.

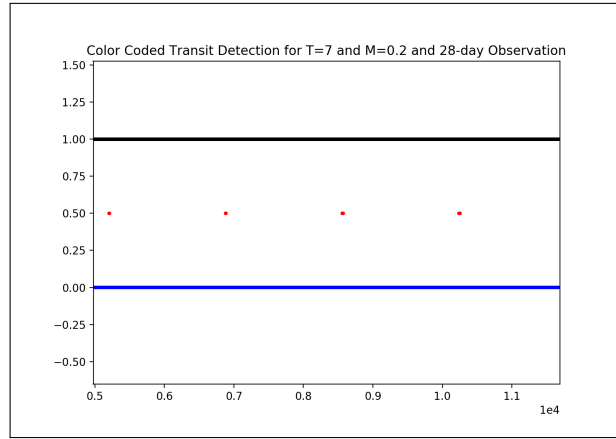


Figure 4.3: SkyHopper visibility = 1 and Transit Function Plot over the whole observational period.

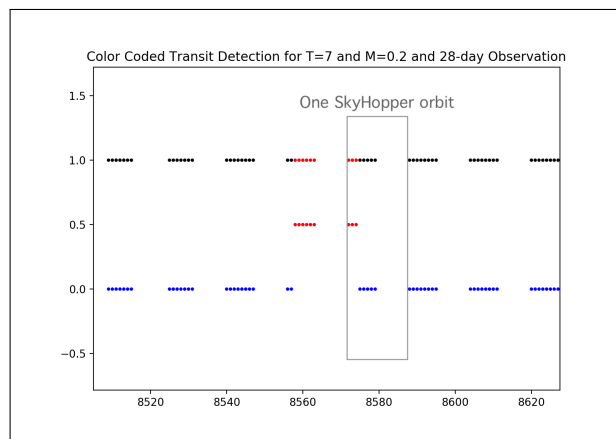


Figure 4.4: Zoomed in version to give a more detailed perspective on what is happening for each coded color.

The plot points are colour coded such that:

- Red $\implies$ Transit detected / Transit steps in PTF
- Black $\implies$ Non-transit detected
- Blue $\implies$ Non-transit steps in PTF

Also note that the plots only show plot points where SkyHopper’s visibility is 1. The visibility = 0 plot points are removed so that it is easier to see the non-transit detections.

4.6 Constraints on Transit Detections

In Figure 4.3, the four small red patches implies that there are four potential transits detected, and Figure 4.4 shows a single transit observed over two SkyHopper orbits around the earth. However, there are conditions that has to be met before the potential detection is claimed to be an actual detection. In this simulation, the passing condition for a detected transit is that during the visible period of each SkyHopper orbit, there must be at least 2 timesteps each of transits and non-transit detected (i.e. each orbit in the combined array must have at least two of each values 1.0 and 1.5).

Such constraints are required because even though SkyHopper may have detected a luminosity drop but with only one time step in the low luminosity regime, one cannot be certain that it has detected a transit. The luminosity drop detected must last long enough to be identified as a transit. On top of that, one also has to ensure that they can see the luminosity dip or rise at the start or end of the planetary transit respectively. Otherwise, if SkyHopper detects only something of low luminosity during its visibility period, it could also potentially be a scaling error or some other instrumental error instead of a planetary transit. Hence, a minimum of two non-transit detection timesteps are required to confirm an actual dip or rise of luminosity.

For example, the transit detected in both the orbits in Figure 4.3 passed this condition, thus SkyHopper can claim transit detection. However, these two transits observed over two SkyHopper orbits are from the same transit in the same planet orbit. To prevent double counting, the transit counter is set to count for only one transit per orbital period of the planet. This is so that even if SkyHopper finds two transits in both of its orbits and they both passed the conditions, only one will be counted.

4.7 Detection Efficiency

Finally, the number of transits detected that passed the conditions, n_{pass} can be used to deduce the efficiency, ϵ of SkyHopper in finding transits:

$$\epsilon = \frac{n_{pass}}{n_{intrinsic}} \quad (8)$$

where the number of intrinsic transits, $n_{intrinsic}$, is the actual number of transits in the PTF that has occurred regardless if they have been spotted by SkyHopper.

This efficiency is essentially the percentage of transits SkyHopper will detect if it were to stare at one specific location in the sky. However, the aim of this project is to investigate the all-sky transit detection efficiency of SkyHopper. In the next section, we will see how this algorithm can be expanded into an all-sky simulation.

5 ALL-SKY SIMULATION

Expanding the single-coordinate simulation into an all-sky simulation is not very difficult, except that we will now be dealing with 3-D arrays instead of a 1-D arrays for SkyHopper visibilities on each coordinate in the sky.

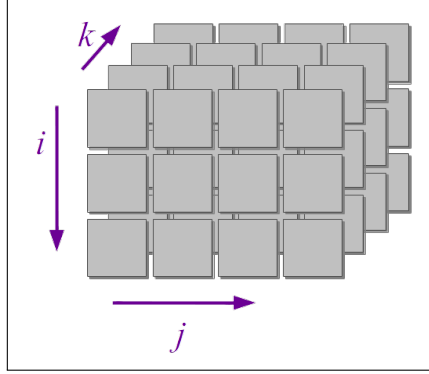


Figure 5.1: $i = \text{Ra}$, $j = \text{Dec}$ and $k = \text{timestep}$. Each square will carry the value of either 0 or 1 which denotes the visibility of SkyHopper. (Image credits to Programming Idioms).

Again, the random observing start timestep generator will provide a random time for the start of SkyHopper’s observation. Then the relevant time frame can be slice out from the 3D year-long all-sky visibility array along the i - j plane, resulting in a $90 \times 180 \times N$ array

The next step is to loop this 3D array over all $[\text{Ra}, \text{Dec}]$ so that every single visibility array of each coordinate can be combined with the randomly provided PTF array to obtain transit detected data.

5.1 SkyHopper’s Transit Detection Efficiency Map

The efficiency of exoplanet transit detection of each coordinate is recorded in a separate 2D array of shape 90×180 . This array can be visualised using a 2D colour map. The Dec and Ra are given on the x and y axis respectively and the colour scale will provide information on the efficiency.

An example is shown in Figure 5.2 on the next page. This plot shows that SkyHopper has detection efficiency of 0.25 almost throughout the whole sky although there are a few regions of exception. Firstly, there are two patches of 0.0 efficiency. Since SkyHopper’s camera cannot be pointed directly towards the sun as it would damage the optical components in SkyHopper, this explains one of the 0.0 efficiency patch. Additionally, SkyHopper’s solar panels have to be constantly exposed to the sun, resulting in an anti-sun region (see Figure 5.3) which SkyHopper will never be able to observe. This explains the second 0.0 efficiency patch.

There are also small regions of higher efficiency (~ 0.5) compared to the rest of the sky. It is suspected that these are the small regions that SkyHopper is able to observe constantly even under it’s rotational constraints, hence no transit detections are missed. However, the efficiency of detection is not 1.0 because the success of detection is still dependent on the constraints imposed in Section 4.6.

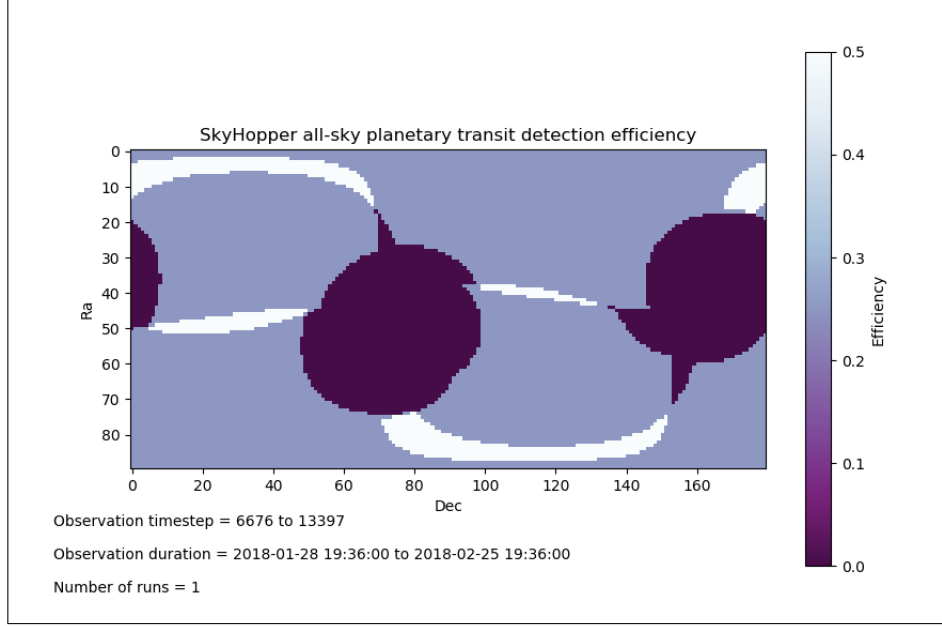


Figure 5.2: SkyHopper’s all-sky exoplanet transit detection efficiency map for $M = 0.2$, $T = 7$ and $tg_days = 28$.

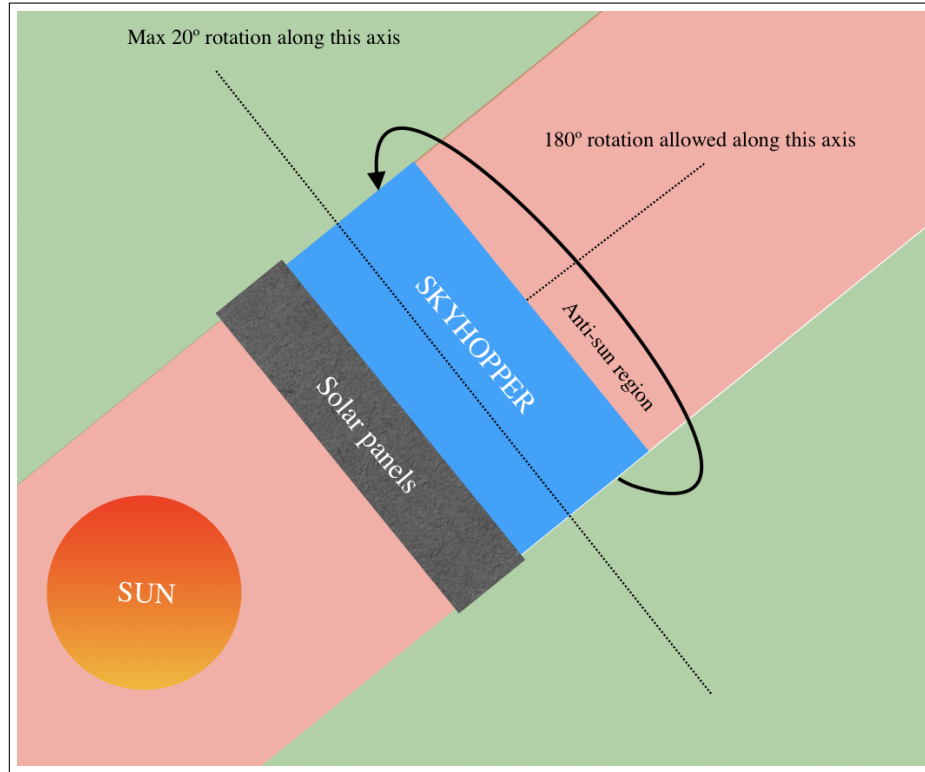


Figure 5.3: Illustration of SkyHopper’s field of view where green is the region within SkyHopper’s visibility and red is the Sun/Anti-sun regions which are not visible to SkyHopper.

5.2 SkyHopper’s All-sky Transit Detection Probability Map

For further analysis, an analogous probability map of SkyHopper detecting transits can also be plotted. If there is at least one successful transit detected by SkyHopper at a single coordinate, the probability is 1, otherwise it is zero. This may seem redundant for a single run, but it will come in handy for a monte carlo simulation.

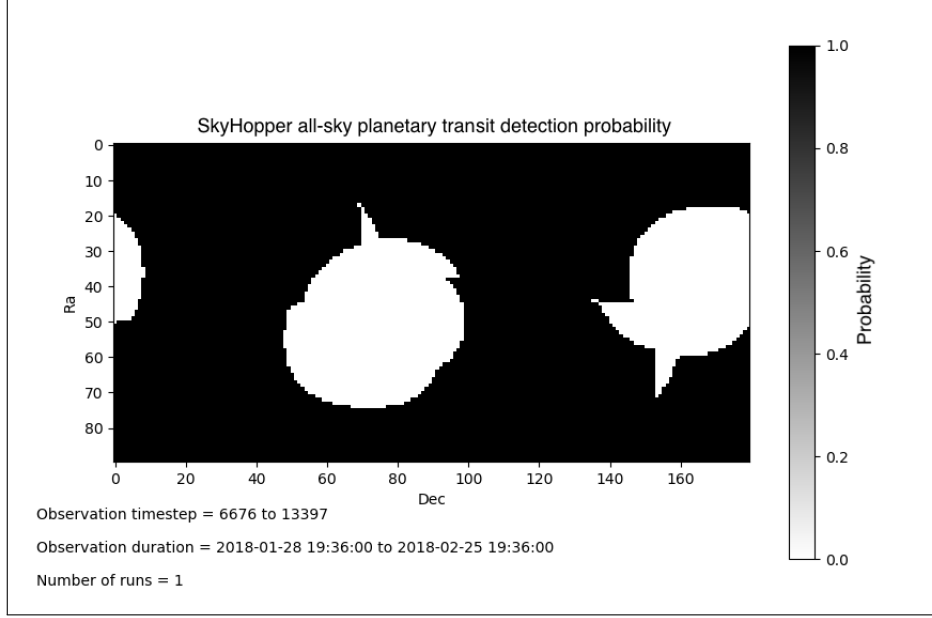


Figure 5.4: SkyHopper’s all-sky exoplanet transit detection probability map.

6 MONTE CARLO SIMULATION

Recall that for given an observational time frame, the location of the planet in the orbit at the start of observation cannot be determined. Hence it is provided randomly by a random number generator. Due to such uncertainty, a Monte Carlo simulation is required. This means to run over the code many times and take the average of all the efficiency maps to obtain the average efficiency of SkyHopper’s transit detection. The run number is precisely the `run_number` input from the parameter file mentioned in Section 3.2.

The same can be done for the probability map which was introduced in Figure 5.4. Having averaged over 60 runs, the probability map becomes useful as it shows the regions of highest probability of successful transit detection.

As expected, with greater number of simulation runs, the detection efficiency and probability across the whole sky becomes more uniform except for regions near the sun and anti-sun patch as the sun moves over time. Another feature that can be spotted here is the trace of lower efficiency and probability extending across the sun and anti-sun patch. This is simply the reduction of detection efficiency and probability due to occultation by the moon as it moves across the sky over SkyHopper’s observation period.

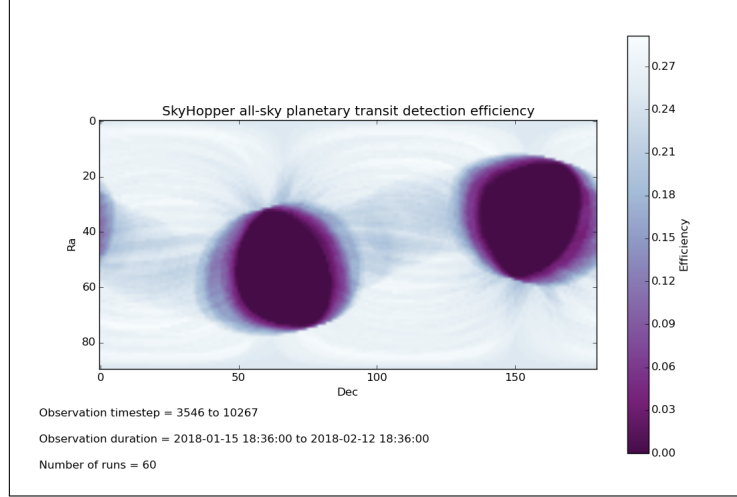


Figure 6.1: Efficiency map for $M = 0.2$, $T = 7$ and $tg_days = 28$ averaged over 60 runs.

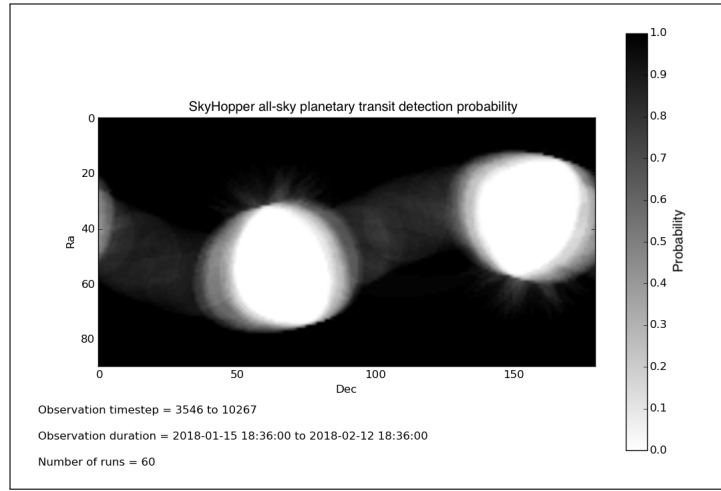


Figure 6.2: Probability map for $M = 0.2$, $T = 7$ and $tg_days = 28$ averaged over 60 runs.

7 SIMULATION SUMMARY

The flowchart on the last page of the report is a general outline of the all-sky Monte Carlo simulation of SkyHopper’s exoplanet transit detection. It summarises the contents discussed in Sections 3, 4, 5 and 6.

8 CONCLUSION

Logically, the efficiency and probability of SkyHopper in detecting planets of different orbital period around host stars of different masses will vary. However, due to time constraint, I was unable to perform Monte Carlo simulations on these various parameters.

In spite of that, this simulation seems to be producing sensible and promising results. Thus, once the Monte Carlo simulations are performed on different sets of parameters (M and T), the results can be used to optimise SkyHopper's search for exoplanets of specific orbital periods hosted by stars with known masses.

In conclusion, the outcome of this science research project progress is a simulation which can contribute towards SkyHopper's mission in search of exoplanet transits in the future upon its tentative launch in 2021^[1].

References

- [1] SkyHopper - For Scientist. <https://skyhopper.research.unimelb.edu.au/for-scientists/>. Accessed: 5 Nov 2018.
- [2] Don Kolinski. HD209458 Transit. <https://www.hao.ucar.edu/research/stare/stare.html>. Accessed: 5 Nov 2018.
- [3] Vera Maria Passegger. The Paradox of Overfitting. Master's thesis, Universität Wein, Universitätsring 1, 1010 Wien, Austria, 2013.

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